
3AL Tutorial 3 2024

1. Prove that G/G is isomorphic to the trivial group.
2. Prove, in two separate ways, that $G/\{1\} \cong G$ (with and without using the First Isomorphism Theorem).
3. Let $T_2(\mathbb{R})$ be the multiplicative group of 2×2 upper triangular matrices with entries from $\mathbb{R}$ with nonzero determinant. Consider the following subset of $T_2(\mathbb{R})$:

$$U = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} : b \in \mathbb{R} \right\}.$$

- (a) Prove that U is a normal subgroup of $T_2(\mathbb{R})$.
 - (b) Show that $U \cong \mathbb{R}$ (where $\mathbb{R}$ is the additive group of real numbers).
 - (c) Use the First Isomorphism Theorem to show that $T_2(\mathbb{R})/U \cong \mathbb{R}^\times \times \mathbb{R}^\times$, where $\mathbb{R}^\times$ is the multiplicative group of nonzero real numbers.
4. The unit circle in the complex plane can be expressed as the set

$$S^1 = \{e^{2i\pi\theta} : \theta \in \mathbb{R}\}.$$

- (a) Prove that S^1 is a subgroup of $\mathbb{C}^\times$.
- (b) Consider the additive group $\mathbb{R}$ with subgroup $\mathbb{Z}$. What do the elements of the factor group $\mathbb{R}/\mathbb{Z}$ look like? Give (at least) two examples.
- (c) Prove that $\mathbb{R}/\mathbb{Z} \cong S^1$.

HINT: Recall that each $z \in \mathbb{C}$ can be written as $z = re^{i\theta}$ for some $r, \theta \in \mathbb{R}$. What does the product of two complex numbers in polar form look like? What is the multiplicative identity?

5. Prove that $\mathbb{R}^\times/\{1, -1\} \cong \mathbb{R}^{>0}$, where $\mathbb{R}^{>0}$ is the multiplicative group of positive real numbers.
6. Prove that $\mathbb{Z}_{18}/\langle [3]_{18} \rangle \cong \mathbb{Z}_3$. (The subscript 18 denotes that the equivalence class is using modulo 18)

Extra problems

(I) Let G_1 and G_2 be groups with $H_1 \triangleleft G_1$ and $H_2 \triangleleft G_2$. Prove the following:

(a) $H_1 \times H_2 \triangleleft G_1 \times G_2$,

(b) $(G_1 \times G_2)/(H_1 \times H_2) \cong (G_1/H_1) \times (G_2/H_2)$.